Jihyung Lee

NLP Researcher | Ph.D. in Artificial Intelligence, POSTECH
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Seoul, Republic of Korea | Korean (native) | English (professional working proficiency)

RESEARCH SUMMARY

NLP researcher working on multi-hop reasoning, retrieval-augmented generation (RAG), and explainable, verifiable question answering, with a focus on the legal domain. Designs multi-stage RAG frameworks, builds domain-specific evaluation benchmarks with domain experts, and develops answer-verification methods that reduce unsupported LLM outputs. Publications at EMNLP and in *Knowledge-Based Systems*; prior work on machine translation and automatic post-editing (WMT 2020). Experienced in delivering research outcomes in government-funded national R&D and industry–academic collaborations.

EDUCATION

Ph.D. in Artificial Intelligence, POSTECH Mar. 2021 – Aug. 2026
Advisor: Prof. Gary Geunbae Lee. Research: legal NLP, multi-hop reasoning, and RAG.

M.S. in Computer Science and Engineering, POSTECH Mar. 2019 – Feb. 2021
Advisor: Prof. Jong-Hyeok Lee. Research: machine translation, with a focus on automatic post-editing (APE).

B.S. in Creative IT Engineering, POSTECH Mar. 2013 – Aug. 2018

PUBLICATIONS

Journal Articles

- **Jihyung Lee**, Gary Geunbae Lee. "Structured Reasoning and Answer Verification: Enhancing Question Answering System Accuracy and Explainability." *Knowledge-Based Systems,* vol. 311, article 113091, 2025. https://doi.org/10.1016/j.knosys.2025.113091
  - Developed an answer verification and correction system that generates entailment trees and reasoning graphs (NLProofS-based stepwise prover with an answer-aware verifier); improved answer Exact Match on EntailmentBank-FF by 1.76 (T5-large) and 3.53 (GPT-3.5) points, also evaluated on three STREET tasks.
  - Built an LLM-based data augmentation pipeline (hypothesis generation, reasoning-chain generation, proof validation) producing 6,000 synthetic training examples from HotpotQA.

Conference Papers

- **Jihyung Lee**[†], Daehui Kim[†], Seonjeong Hwang, Hyounghun Kim, Gary Geunbae Lee. "KoBLEX: Open Legal Question Answering with Multi-hop Reasoning." In *Proceedings of EMNLP 2025*, pp. 4019–4053, Suzhou, China, 2025. ([†]Equal contribution)
  Paper: https://aclanthology.org/2025.emnlp-main.200/ | Code: https://github.com/continu2720/KoBLEX
  - Led data generation and quality assurance for a provision-grounded legal QA benchmark of 226 scenario-based multi-hop instances across 83 Korean statutes.
  - Generated and curated data with an LLM-based pipeline (incremental multi-hop expansion, scenario-based reformulation); worked with Korean law-school graduates to turn expert feedback into explicit quality criteria (evidence requirements, scenario–question–answer consistency, legal accuracy), applied in a two-stage process of automated LLM checks followed by expert revision and evaluation.

Preprints

- **Jihyung Lee**, Hyounghun Kim, Gary Geunbae Lee. "Decompose-and-Refine: Structured Legal Question Answering with Parametric Retrieval." *arXiv preprint,* 2026. https://arxiv.org/abs/2605.24454
  - Proposed Decompose-and-Refine (DAR), a multi-hop RAG framework that bridges the vocabulary gap between lay user questions and statutory text by combining step-wise question decomposition with multi-channel query refinement.
  - Designed an evidence-grounded pipeline of BM25 retrieval, reciprocal rank fusion, cross-encoder reranking, and LLM-based selection of one core provision per sub-question, aggregating selected provisions to ground final answer generation.
  - Without task-specific LLM training, improved retrieval F1 by 12.39 and LF-Eval by 12.13 points over

ParSeR on KoBLEX (Qwen3-32B); also evaluated with Gemma3-27B.

Workshop Papers

- **Jihyung Lee**, WonKee Lee, Jaehun Shin, Baikjin Jung, Young-Kil Kim, Jong-Hyeok Lee. "POSTECH-ETRI's Submission to the WMT2020 APE Shared Task: Automatic Post-Editing with Cross-lingual Language Model." In *Proceedings of the Fifth Conference on Machine Translation* (*WMT 2020*), pp. 777–782, 2020. https://aclanthology.org/2020.wmt-1.82
  - Built cross-lingual Transformer-based APE systems for English–German and English–Chinese using masked and translation language modeling objectives to learn joint source–target representations.
  - Combined large-scale synthetic APE triplets with task-specific fine-tuning and ensembling to address limited human post-editing data; improved over raw MT output by +6.16 BLEU / −4.54 TER (En–De) and +5.78 BLEU / −4.57 TER (En–Zh).
- WonKee Lee, Jaehun Shin, Baikjin Jung, **Jihyung Lee**, Jong-Hyeok Lee. "Noising Scheme for Data Augmentation in Automatic Post-Editing." In *Proceedings of WMT 2020*, 2020. https://aclanthology.org/2020.wmt-1.83

Domestic Conference Papers

- **이지형**, 이근배. "질의응답 결과 재순위화를 위한 자연어 추론 모델." 제33회 한글 및 한국어 정보처리 학술대회 논문집, pp. 405–409, 2021. https://koreascience.kr/article/CFKO202130060642819.page
- **이지형**, 이원기, 김영길, 이종혁. "번역문 사후교정을 위한 교차언어 모델 기반의 전이학습." 한국정보과학회 학술발표논문집, pp. 392–394, 2020. https://dbpia.co.kr/journal/articleDetail?nodeId=NODE09874449

RESEARCH PROJECTS

High-Performance Knowledge System Development Project
Government-funded national R&D, Ministry of Science and ICT (MSIT), Korea — Mar. 2024 – Nov. 2025

- Developed evidence-grounded, multi-document QA by integrating LLMs with RAG; evaluated the same pipeline on commercial and open-weight models, improving Exact Match over the LLM-only (no-retrieval) baseline from 0.47 to 0.64 (GPT-4o-mini) and from 0.32 to 0.54 (LLaMA).
- Developed explainable QA that outputs structured reasoning graphs alongside answers, so each answer's reasoning can be inspected step by step.
- As student lead for Subproject 2 (of three; PI: Prof. Gary Geunbae Lee), compiled the 2025 multi-year stage report to the funding agency, consolidating results from POSTECH, Seoul National University, and Dongguk University across project years; audited prior reports, reconciled inconsistent figures, and recovered missing records.

Artificial Intelligence Technology for Inferring Issues and Supporting Facts from Raw Text
Government-funded national R&D, MSIT, Korea — Apr. 2021 – Dec. 2022

- Designed a verification layer for a QA system: a Transformer cross-encoder that reranks candidate answers from extractive QA, summarization, and graph-based generation, plus an ELECTRA-based NLI model that checks whether each answer is entailed by its evidence.
- Achieved 86.79% answer-verification accuracy on HotpotQA against an 85% project target; passed official third-party performance testing by the Telecommunications Technology Association (TTA).
- Deployed the reranker as a Dockerized Flask REST API with GPU inference, routing requests to English, Korean common-sense, and Korean legal models; integrated with other institutions' modules via a shared JSON API. Code: https://github.com/continu2720/Fact-Reasoning-POSTECH

Industry–Academic Collaborative Research — Explainable Legal QA
Company L — May 2024 – Jun. 2024

- Developed a multi-hop machine reading comprehension system for explainable legal QA using fine-tuned Transformer-based language models.
- Delivered within a two-month timeline without task-specific legal QA data by staging results: first on science-domain multi-hop QA data, then on public legal datasets from AI Hub (MSIT); presented final results on-site to the partner.

Industry–Academic Collaborative Research — LLM Output Reliability
Company S Aug. 2023 – Dec. 2023

- Researched LLM-based answer verification and data augmentation methods to improve the reliability and filtering of generated outputs (details under NDA); presented final results on-site to the partner's team leads.

PATENTS & SOFTWARE REGISTRATION

- **Jihyung Lee**, Gary Geunbae Lee. "Apparatus and Method for Re-ranking Question Answering Results Using a Natural Language Inference Neural Network Model." Korean Patent Application No. 10-2021-0151973, filed Nov. 8, 2021.
- **Jihyung Lee**. "Question Answering Result Re-ranking System." Software Registration No. C-2021-037012, registered Sep. 3, 2021.

TECHNICAL SKILLS

Programming: Python (primary), Bash, C++
ML & LLM: PyTorch, Hugging Face Transformers, vLLM, open-weight LLMs (LLaMA, Qwen, Gemma), Claude API, OpenAI API, OpenRouter
Methods: Multi-hop RAG, query refinement, hybrid retrieval and reranking, NLI-based answer verification, LLM-as-judge evaluation, synthetic data generation, benchmark construction
Infrastructure: Linux, Git, Slurm (multi-GPU cluster scheduling), Docker, Flask REST APIs, SQLite, Oracle Cloud Infrastructure, Tailscale, Cloudflare (DNS, Tunnel)

TEACHING & LEADERSHIP

- **Teaching Assistant, NLP Course** — POSCO Apr. 2025
  Supported about 70 university students, explaining NLP concepts and practical AI applications to learners from non-research backgrounds.
- **Student Lead, NLP Group** — POSTECH Sep. 2022 – Aug. 2023
  Managed student affairs and coordinated administrative activities within the POSTECH NLP Group.

INDEPENDENT PROJECTS

Self-Hosted LLM Agent & Personal Infrastructure 2025 – present
Personal project | Oracle Cloud (ARM) server, Docker, Tailscale, Cloudflare

- Migrated Hermes Agent (open-source LLM agent framework) from a PC to a cloud server, running it as a Discord bot with scheduled jobs (e.g., a daily NLP job-posting briefing from my own web crawler) under a dedicated unprivileged account with systemd and Tailscale-only access; also compare models via OpenRouter for quality, cost, and task fit.
- Built a persistent job queue (Python, SQLite) that passes tasks from the server-side agent to an app that runs only on my PC, with leases, retries with exponential backoff, and token authentication; found and fixed an outage caused by unclosed SQLite connections.

EMPLOYMENT

Game Developer — LINE Games, Seoul (full-time, on leave of absence from B.S.) Aug. 2017 – Feb. 2018

- Launched a live-service rhythm game (*Protocol: Hyperspace Diver*) in a cross-functional team spanning programming, art direction, and production; sourced and contracted external illustrators for song cover art.

CREATIVE ACTIVITY

POSTECH Game Development Club "G-POS" 2013 – 2018

- **Five Ani** (solo developer): original puzzle game—gameplay mechanics, programming, and design.
- **WeNi** (UI designer): UI assets and in-game visual components for an Android RPG.
- **Doubling** (game designer / programmer / artist): Unity-based puzzle game.